

BeautyGlow: On-Demand Makeup Transfer Framework with Reversible Generative Network

Hung-Jen Chen, Ka-Ming Hui, Szu-Yu Wang, Li-Wu Tsao, Hong-Han Shuai, Wen-Huang Cheng
 National Chiao Tung University, Taiwan

{hjc.eed07g, ming2242.cm06g, joey840404.eed03, leo159753.me05, hhshuai, whcheng}@nctu.edu.tw

Abstract

As makeup has been widely-adopted for beautification, finding suitable makeup by virtual makeup applications becomes popular. Therefore, a recent line of studies proposes to transfer the makeup from a given reference makeup image to the source non-makeup one. However, it is still challenging due to the massive number of makeup combinations. To facilitate on-demand makeup transfer, in this work, we propose BeautyGlow that decompose the latent vectors of face images derived from the Glow model into makeup and non-makeup latent vectors. Since there is no paired dataset, we formulate a new loss function to guide the decomposition. Afterward, the non-makeup latent vector of a source image and makeup latent vector of a reference image are effectively combined and revert back to the image domain to derive the results. Experimental results show that the transfer quality of BeautyGlow is comparable to the state-of-the-art methods, while the unique ability to manipulate latent vectors allows BeautyGlow to realize on-demand makeup transfer.

1. Introduction

As the slogan says, “Life isn’t perfect, but your makeup can be.” Nowadays, makeup provides a convenient way to improve the facial appearance, e.g., powder foundations for hiding the skin imperfections, blushes for creating chubby cheeks. Although makeup is ubiquitous in daily life, it is not easy to find the best-suited makeup because 1) makeup trials are time-consuming and inconvenient, and 2) there are thousands and thousands of cosmetic products, varying from brands, functions, colors, and way-to-use, which leads to a more intractable number of the makeup style combinations. To address the first issue, virtual makeup applications that provide beautification function come up to facilitate the makeup trials, such as YoucamMakeup¹ and Meitu². How-

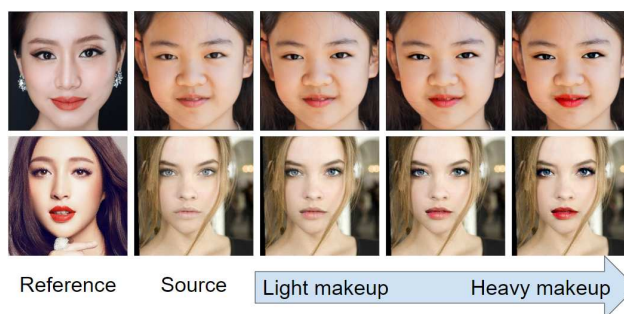


Figure 1: The makeup features such as eyeshadows and lip gloss are extracted from reference makeup images and are transferred to source non-makeup images. The lightness of the makeup can be tuned by adjusting the magnification in the latent space.

ever, how to find suitable makeup for each user is still an open problem.

One of the possible solutions is to let users select the makeup styles from the reference photos of celebrities or friends and transfer the makeup to users’ faces. For example, Guo *et al.* [10] propose to decompose the images into three layers, i.e., face structure layer, skin detail layer, and color layer, and transfer the information from each layer of reference images to the corresponding layer of the target images. However, the predefined layers and transfer function are not data-driven and thus are inclined to generate artifacts in many cases. To directly learn from the data, a dataset of photo triplets (user face, reference makeup, and transferred results) is desirable for learning. Nevertheless, most of the existing datasets only provide the images with makeup and non-makeup pairs. To avoid the need for such triplet datasets, [23] proposes to use CNN for identifying different makeup functions and extracting the corresponding features. Afterward, different loss functions are defined for different parts of makeup to make the after-makeup images natural, e.g., foundations are transferred by regularizing the inner product of the feature maps for smoothing the

¹<https://www.perfectcorp.com/app/ymk>

²<http://makeup.meitu.com/en/>

BeautyGlow: 基于可逆生成网络的按需美妆迁移框架

陈宏仁, 许家铭, 王思宇, 曹立武, 柴鸿涵, 郑文煌 国立交通大学, 台湾

{hjc.eed07g, ming2242.cm06g, joey840404.eed03, leo159753.me05, hhshuai, whcheng}@nctu.edu.tw

摘要

As makeup has been widely-adopted for beautification, finding suitable makeup by virtual makeup applications becomes popular. Therefore, a recent line of studies proposes to transfer the makeup from a given reference makeup image to the source non-makeup one. However, it is still challenging due to the massive number of makeup combinations. To facilitate on-demand makeup transfer, in this work, we propose BeautyGlow that decompose the latent vectors of face images derived from the Glow model into makeup and non-makeup latent vectors. Since there is no paired dataset, we formulate a new loss function to guide the decomposition. Afterward, the non-makeup latent vector of a source image and makeup latent vector of a reference image are effectively combined and revert back to the image domain to derive the results. Experimental results show that the transfer quality of BeautyGlow is comparable to the state-of-the-art methods, while the unique ability to manipulate latent vectors allows BeautyGlow to realize on-demand makeup transfer.

1. 引言

正如口号所说, “生活不完美, 但妆容可以。”如今, 化妆提供了一种方便的方式来改善面部外观, 例如使用粉底遮盖皮肤瑕疵, 用腮红打造丰盈脸颊。尽管化妆品在日常生活中无处不在, 但找到最适合自己的化妆品并不容易, 原因有二: 一是试妆耗时且不便; 二是市面上的化妆品种类繁多, 从品牌、功能、颜色到使用方式各不相同, 导致化妆风格组合的选择更加复杂多样。为了解决第一个问题, 提供美化功能的虚拟化妆应用程序应运而生, 以方便用户进行试妆, 例如YoucamMakeup和Meitu等。如何

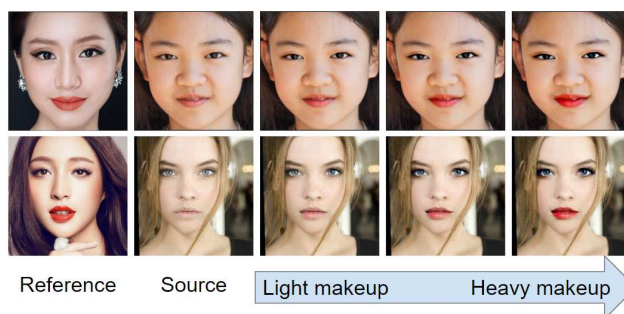


图1: The makeup features such as eyeshadows and lip gloss are extracted from reference makeup images and are transferred to source non-makeup images. The lightness of the makeup can be tuned by adjusting the magnification in the latent space.

始终如何为每位用户找到合适的化妆品仍然是一个开放性问题。

一种可能的解决方案是让用户从名人或朋友的照片中选择妆容风格, 并将这些妆容转移到用户的脸部。例如, Guo *et al.* [10] 提出将图像分解为三层, 即面部结构层、皮肤细节层和色彩层, 并将参考照片中的每一层信息传输到目标图像的相应层。然而, 预定义的层次和传输函数并非数据驱动, 在许多情况下容易生成伪影。为了直接从数据中学习, 需要一个包含三元组 (用户脸部、参考妆容及转移结果) 的数据集进行训练。不过, 现有的大多数数据集仅提供带有妆容与无妆容的照片对。为了避免使用这种三元组数据集, [23] 提出利用 CNN 识别不同的妆容功能并提取相应的特征。之后, 为化妆的不同部分定义了不同的损失函数, 以使化后的图像看起来更自然, 例如, 通过正则化特征图的内积来平滑底妆转移。

¹<https://www.perfectcorp.com/app/ymk>

²<http://makeup.meitu.com/en/>

skin’s texture. Although previous work generates good results on makeup transfer, the domain knowledge is required to design different functions to generate different makeup.

Recent years, Generative Adversarial Networks (GANs) are widely used for generating high-resolution realistic images. For style transfer on whole images (e.g., painting style), [30] proposes CycleGAN to incorporate the cycle consistency loss to train the mapping function between two domains by adopting two generators and discriminators. However, CycleGAN can only generate a general makeup style rather than a specific makeup style. Based on CycleGAN, Chang *et al.* [1] introduce an asymmetric function and train a makeup removal and transfer network together to preserve the face identity. Moreover, [19] utilizes the GAN framework and further proposes pixel-level histogram loss to maximize the similarity of makeup style, while perceptual loss and cycle consistency loss are designed to preserve identity. Nonetheless, there is no encoder in GAN-based methods and thus cannot adjust the makeup extent by interpolating the latent space. Adjusting the makeup from light to heavy is important for finding the best-suited makeup since users can obtain many candidates from one reference images.

A recent line of studies synthesizes images by flow-based generative models, of which the encoder function is reversible and supports different applications by manipulating the latent space and reserving it back to real image space [4, 15]. Due to the ability of manipulating latent space, in this paper, we propose an unsupervised on-demand makeup transfer approach, namely, BeautyGlow, based on the Glow architecture [15]. Specifically, Glow provides a general framework to learn a generative network with invertible functions that encode input images into a meaningful latent space and enable modification of existent data points. Although Glow allows manipulating the latent vectors, it is challenging to transfer the local makeup details of the reference image to the target image since the makeup and face are mixed as the latent vector. One of the possible approaches is to find the average latent vector of makeup images and average latent vector of non-makeup images, and then use the difference as the direction of manipulating. However, this approach contains two major issues: 1) it can only find the general makeup but not the user-specified makeup, and 2) it requires a lot of images from the same person or same makeup to find the correct average latent vector.

To address this issue, BeautyGlow first defines a transformation matrix that decomposes the latent vectors into latent vector of makeup features and latent vector of facial identity features. However, due to the lack of paired data, we further formulate a new loss function containing perceptual loss, makeup loss, intra-domain loss, inter-domain loss, and cycle consistency loss, to guide the decomposi-

tion. Compared with other methods based on GANs, BeautyGlow does not need to train two large networks, i.e., generator and discriminator, which makes it more stable. Most importantly, the invertible meaningful latent space with the transformation matrix facilitated the on-demand makeup transfer, that is, users can freely adjusting the makeup from light to heavy and BeautyGlow spontaneously synthesizes the results (less than 1 second).³ The contributions of this paper are summarized as follows.

- Inspired by Glow, we propose BeautyGlow that can transfer the makeup from reference image to target image. The meaningful latent space facilitates on-demand makeup adjustment. To the best of our knowledge, this is the first Glow-based makeup transfer framework.
- New transformation matrix and loss function are formulated to guide the model training. It is worth noting that the proposed framework can be easily extended to other applications that require decomposing the latent image vector into two latent vectors, e.g., rain removal, fog removal.
- Experimental results on quantitative and qualitative comparison manifest that the proposed BeautyGlow is comparable to the state-of-the-art methods, while the manipulation on latent vectors can generate realistic images from light makeup to heavy makeup.

2. Related works

2.1. Makeup Studies

Traditionally, image processing techniques are applied for makeup transfer. For example, image analogy [11] introduces a framework that requires a pair of well-aligned before-makeup and after-makeup photos of the same person for makeup transfer. Guo *et al.* decompose facial details by decomposing reference images into three layers and transfer information from each layer to the corresponding layer of the other image [10]. Moreover, [28] proposes to detect the face landmarks and transfer the makeup by adjusting the landmark with skin color GMM-based segmentation. In addition to process the makeup, [17] manipulates the intrinsic image layers with a physically-based reflectance model to simulate the makeup with different lighting conditions in a photo. Compared with these work, the proposed BeautyGlow can perform makeup transfer and makeup removal simultaneously with on-demand makeup adjustment by manipulating the latent space without the need of post-processing.

³A demo is available at <https://beautyglow.github.io/>.

皮肤的纹理。尽管之前的工作在化妆品迁移上取得了良好的结果，但需要特定领域的知识来设计不同的功能以生成不同类型的妆容。

近年来，生成对抗网络（GAN）被广泛用于生成高分辨率的逼真图像。对于整个图像上的风格迁移（如绘画风格），文献[30]提出了CycleGAN，通过采用两个生成器和判别器来引入循环一致性损失以训练两个域之间的映射函数。然而，CycleGAN只能生成通用的化妆风格而无法生成特定的化妆风格。基于CycleGAN，Chang *et al.* [1] 引入了一个不对称函数，并共同训练了去除和转移妆容的网络，以保持面部身份特征。此外，文献[19]利用GAN框架进一步提出了像素级直方图损失，以最大化化妆风格的相似性，同时感知损失和循环一致性损失被设计用于保留身份信息。然而，在基于GAN的方法中没有编码器，因此无法通过插值潜在空间来调整妆容的程度。从轻到重地调整妆容对于找到最适合用户的妆容非常重要，因为用户可以从一张参考图像中获得多个候选方案。

近期的研究通过基于流的生成模型综合图像，其中编码函数是可逆的，并且可以通过操纵潜在空间并将其恢复到真实图像空间来支持不同的应用[4, 15]。由于能够操纵潜在空间的能力，在本文中，我们提出了一种无监督按需美妆迁移方法，即BeautyGlow，基于Glow架构[15]。具体来说，Glow提供了一个通用框架，用于学习具有可逆函数的生成网络，该网络将输入图像编码为有意义的潜在空间，并使现有数据点得以修改。尽管Glow允许操纵潜在向量，但由于妆容和人脸混合在潜在向量中，因此很难将参考图像中的局部美妆细节转移到目标图像上。一种可能的方法是找到化妆图像的平均潜在向量和非化妆图像的平均潜在向量，然后使用两者的差异作为操作的方向。然而，这种方法存在两个主要问题：1）它只能找到通用妆容而不能满足用户指定的妆容需求；2）需要大量来自同一个人或相同妆容的照片才能找到正确的平均潜在向量。

为了解决这一问题，BeautyGlow 首先定义了一个转换矩阵，将潜在向量分解为妆容特征的潜在向量和面部身份特征的潜在向量。然而，由于缺乏配对数据，我们进一步提出一个新的损失函数，该函数包含感知损失、妆容损失、域内损失、域间损失以及循环一致性损失，以指导这一分解过程。

与基于GAN的其他方法相比，BeautyGlow无需训练两个大型网络（即生成器和判别器），因此更加稳定。最重要的是，可逆且具有意义的潜在空间以及转换矩阵使得按需化妆转移成为可能，用户可以自由地调整从淡妆到浓妆的程度，而BeautyGlow会自动生成结果（少于1秒）。本文的主要贡献总结如下。

- 受Glow的启发，我们提出了BeautyGlow，它可以将参考图像中的妆容转移到目标图像上。有意义的潜在空间便于按需调整妆容。据我们所知，这是首个基于Glow的妆容转移框架。
- 新的变换矩阵和损失函数被设计出来以指导模型训练。值得注意的是，所提出的框架可以轻松扩展到其他需要将潜在图像向量分解为两个潜在向量的应用中，例如去除雨纹、消除雾气。
- 实验结果表明，无论是定量还是定性比较，所提出的BeautyGlow方法均与当前最先进的方法相当。通过对潜在向量的操作，可以从淡妆生成到浓妆的逼真图像。

2. 相关工作

2.1. 构造研究

传统上，图像处理技术被应用于妆容转移。例如，图像类比法[11]提出了一种框架，该框架要求使用同一人的两张对齐良好的未化妆和已化妆的照片来进行妆容转移。郭某等人将参考图像分解为三层，并从每一层传输信息到另一张图片的相应层次[10]。此外，[28]提出通过调整基于肤色高斯混合模型（GMM）分割后的面部特征点来检测面部标志并进行妆容转移。除了处理妆容之外，[17]还利用物理反射模型操作图像的基本层，在不同光照条件下模拟照片中的妆容效果。与这些工作相比，所提出的 BeautyGlow 可以同时执行妆容转移和去除，并通过操纵潜在空间实现按需调整妆容，而无需进行后处理。

³A demo is available at <https://beautyglow.github.io/>.

2.2. Style Transfer

To mix the content and style together, most works [7, 29] use different layers in CNN to represent styles and content and swap the style for style transfer. The style transfer can be categorized into three classes. 1) Per-Style-Per-Model (PSPM) [14, 25, 26, 8, 21, 20]. PSPM requires training different models for different styles. As the number of styles becomes large (e.g., makeup), it is difficult and inconvenient to use PSPM for style transfer. 2) Multiple-Style-Per-Model (MSPM) [5, 2, 6]. MSPM only requires a trained model to transfer multiple styles by slightly adjusting the instance normalization layers. 3) Arbitrary-Style-Per-Model (ASPM) [13, 9, 12]. ASPM focuses on the zero-shot style transfer, which can transfer the new styles (unseen in the training data) by finding the transformation function via a style prediction network. It is worth noting that the proposed BeautyGlow belongs to ASPM. However, compared with style transfer, the requirement of the details for makeup transfer is much higher than that for style transfer.

2.3. GAN for Style/Makeup Transfer

Due to the good performance of generating realistic images by adversarial network, GAN is widely-used for image-to-image translation. For example, Taigman *et al.* propose a method to transfer a realistic photo to comic characters by GAN and VAE. Moreover, [18, 24, 16, 22] aim to transfer style, attributes or specific details by exploiting GANs. CycleGAN further introduces two coupled generator and a cycle consistency loss to transfer images between two different domains [30], which avoids the requirement of input-output pairs. Although CycleGAN can be directly exploited to makeup transfer, i.e., one generator transfers a general makeup to non-makeup faces, and the other generator removes makeup from makeup faces, it cannot transfer a user-specified style. To fulfill the need to transfer a specific makeup to a non-makeup face, [1] alters the coupled generator in CycleGAN by using two images (source and target) for the makeup generator. The generator generates a makeup mask referenced from the target image and covers it on the source image to complete the makeup transfer. On the other hand, BeautyGAN adopts similar idea with dual input and output for makeup transfer and removal and enhance the correctness of instance-level makeup transfer by matching the color histogram in different segments of the face [19]. However, GAN-based methods contain no encoder to construct the latent space from the data and thus can not realize on-demand makeup transfer by simply interpolating the latent vectors.

2.4. Latent Space Adaption

The likelihood-based approaches, such as Variational AutoEncoders (VAE) and Glow, generate realistic images with a latent space, while manipulating the latent space

model can facilitate developing new image applications, e.g., image editing [31], avatar synthesis [27]. Specifically, VAE optimizes the likelihood by maximizing the variational lower bound and learns a latent space that represents all the data points. On the other hand, flow-based generative models [3, 4] construct a bijective reversible transfer function F from the image space to the latent space so that modifying latent vectors can be mapped back to real images with F^{-1} . Glow further improves the model architecture by a new permutation layer which is learned after updating the model rather than the fixed permutation layer [15].

3. BeautyGlow

3.1. Glow

We first present the Glow framework [15] as the preliminary. Glow introduces an invertible function f comprising a sequence of transformation matrices for mapping images into a latent space, where f is required to fulfill the following properties. (1) The determinant of the Jacobian matrix of the transformation matrix should be calculated efficiently. (2) From the sampled datapoint z in latent space, mapping it back to the data x by using the inverse transformation $x = f^{-1}(z)$ should also be obtained easily. To accomplish these goals, Glow utilizes the additive layer and affine coupling layer introduced in [4]. A coupling layer represents a bijection function, which can update part of the input vector or latent vectors efficiently. Also, squeezing layer is adapted to implement the multi-scale architecture, which divides the image into sub-images of shape $2 \times 2 \times c$, then reshapes them into $1 \times 1 \times 4c$ for squeezing image information into channels. On top of that, before propagating the output of the current level to the next level, half of the dimensions of the output are factored out and dumped into latent space to reduce the computational cost and the number of parameters. Furthermore, Glow replaces the fixed permutation in [4] with a new layer, Invertible 1×1 conv, to avoid some components of the input vector is left unchanged in the coupling layer. Based on this model, we are able to transform input images into meaningful vectors in latent space and manipulate the vectors for on-demand makeup transfer.

3.2. Formulation

Based on the latent space derived from Glow, the goal is to extract the makeup features from the reference makeup image and apply it to the source non-makeup image. Specifically, considering two domains, the non-makeup images domain denoted as $X \subset \mathbb{R}^{h \times w \times c}$ where h , w , and c represent the height of input images, the width of input images, and the number of RGB channels, respectively. The makeup images domain denoted as $Y \subset \mathbb{R}^{h \times w \times c}$, which are encoded into the latent space denoted as $Z \subset \mathbb{R}^{c \times h \times w}$

2.2. 风格转换

为了将内容和风格混合在一起，大多数作品[7, 29]使用CNN的不同层次来表示风格和-content，并交换风格以实现风格迁移。风格迁移可以分为三类。1) 每种风格每种模型 (PSPM) [14, 25, 26, 8, 21, 20]。PSPM要求为不同的风格训练不同的模型。当风格数量增多时 (例如化妆)，使用PSPM进行风格迁移变得困难且不便。2) 每种模型多种风格 (MSPM) [5, 2, 6]。MSPM只需要通过微调实例归一化层即可用一个预训练的模型传输多种风格。3) 每种模型任意风格 (ASPM) [13, 9, 12]。ASPM专注于零样本风格迁移，可以通过样式预测网络找到变换函数来传输新的未见过的样式。值得注意的是，所提出的BeautyGlow属于ASPM类别。然而，与风格迁移相比，化妆迁移对细节的要求远高于一般风格迁移。

2.3. 风格/妆容迁移的GAN

由于对抗网络生成逼真图像的良好性能，GAN被广泛应用于图像到图像的转换。例如，Taigman等人提出了一种方法，通过使用GAN和VAE将真实照片转化为漫画角色形象。此外，[18, 24, 16, 22]旨在利用GAN来转移风格、属性或特定细节。CycleGAN进一步引入了两个耦合生成器和一个循环一致性损失函数，用于在两个不同领域之间转换图像[30]，从而避免了输入输出对的需求。尽管CycleGAN可以直接应用于妆容转换，即一个生成器将一般的妆容转移到无妆脸上，另一个生成器则从有妆脸上移除妆容，但它无法转移用户指定的风格。为了满足特定妆容向无妆脸转换的需求，[1]通过使用两个图像 (源图和目标图) 来修改CycleGAN中的耦合生成器，使得生成器根据目标图像生成一个妆容掩模，并将其覆盖在源图像上以完成妆容转移。另一方面，BeautyGAN采用类似的方法进行双输入输出的妆容转换与移除，并通过匹配面部不同区域的颜色直方图来增强实例级妆容转换的准确性[19]。然而，基于GAN的方法没有编码器从数据中构建潜在空间，因此无法仅通过插值潜在向量实现按需妆容转移。

2.4. 隐空间适配

基于似然的方法，如变分自编码器 (VAE) 和Glow，通过操纵隐空间生成逼真的图像。

模型可以促进开发新的图像应用，例如图像编辑[31]、虚拟形象合成[27]。具体而言，变分自编码器 (VAE) 通过最大化变分下界来优化似然性，并学习一个表示所有数据点的潜在空间。另一方面，基于流的生成模型[3, 4]构建了一个从图像空间到潜在空间的一一对应可逆转换函数 F ，使得修改潜在向量可以映射回真实的图像 F^{-1} 。Glow通过引入一个新的置换层进一步改进了模型架构，该层在更新模型之后进行学习而不是使用固定的置换层[15]。

3. 璀璨光辉

3.1. Glow

我们首先介绍Glow框架[15]作为基础。Glow引入了一个可逆函数 f ，该函数由一系列变换矩阵组成，用于将图像映射到潜在空间中，其中 f 需要满足以下性质：(1) 变换矩阵的雅可比矩阵行列式应能高效计算。(2) 从潜在空间中的采样数据点 z 出发，通过使用逆变换 $x = f^{-1}(z)$ 将其映回原始数据 x 也应易于实现。为了达到这些目标，Glow利用了[4]中引入的加性层和仿射耦合层。耦合层表示一个双射函数，可以高效地更新输入向量或潜在向量的一部分。此外，挤压层被采用以实现多尺度架构，该架构将图像划分为 $2 \times 2 \times c$ 的子图，然后将其重塑为 $1 \times 1 \times 4c$ 形状，以便将图像信息压缩到通道中。在此基础上，在当前层级的输出传递给下一层级之前，会分解出一半维度并将其丢弃到潜在空间中以减少计算成本和参数数量。此外，Glow用新的Invertible 1×1 conv层替换了[4]中的固定置换，以避免输入向量的一部分在耦合层中保持不变。基于此模型，我们可以将输入图像转换为潜在空间中有意义的向量，并操作这些向量进行按需化妆转移。

3.2. 模型表述

基于Glow生成的潜在空间，目标是从参考化妆图像中提取化妆特征并将其应用于原始未化妆图像。具体而言，在两个领域内进行考虑：未化妆图像域表示为 $X \subset \mathbb{R}^{h \times w \times c}$ ，其中 h 、 w 和 c 分别代表输入图像的高度、宽度以及RGB通道的数量；化妆图像域表示为 $Y \subset \mathbb{R}^{h \times w \times c}$ ，这些图像被编码到潜在空间中，该潜在空间表示为 $Z \subset \mathbb{R}^{c \times h \times w}$ 。

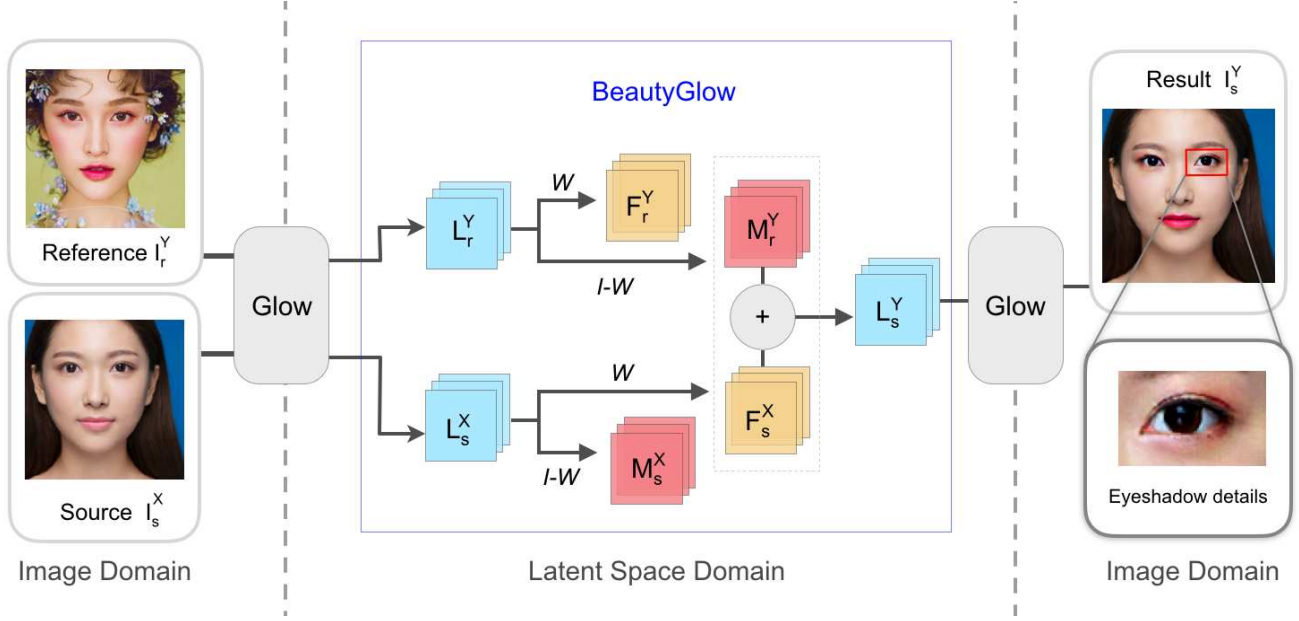


Figure 2: The framework of the BeautyGlow. Glow transforms all images in makeup domain I^Y and all images in the non-makeup domain I^X as inputs: they are encoded into latent space Z , denoted as L^X and L^Y respectively. a vector obtained from L^X as source non-makeup, L_s^X , and a vector obtained from L^Y as reference makeup image, L_r^Y , are decomposed into two latent features (F_s^X, M_s^X) , (F_r^Y, M_r^Y) respectively through a transformation matrix W . L_s^Y is generated by adding F_s^X and M_r^Y together and L_s^Y can also be decomposed into reconstruction latent features, (F_s^Y, M_s^Y) .

through Glow[15].

Given two images as inputs: a source image $I_s^X \in X$ and a reference image $I_r^Y \in Y$ are encoded into two latent vectors, $L_s^X \in Z$ and $L_r^Y \in Z$ respectively with Glow. After that, a transformation matrix W is expected to extract facial features denoted as $F_s^X = L_s^X \cdot W$ and extract makeup features denoted as $M_r^Y = L_r^Y \cdot (I - W)$, where I is identity matrix. Here, we adopt the additive formulation, i.e., adding F_s^X and M_r^Y to form after-makeup latent vector $L_s^Y \in Z$, since 1) additive formulation can be computed on-the-fly, and 2) the interpolation in additive formulation is intuitive. The after-makeup vector L_s^Y is then decoded back to image domain as the after-makeup image $I_s^Y \in Y$. The model architecture of BeautyGlow is shown in Figure 2.

However, how to learn the transformation matrix W that perfectly decomposes the latent vectors into latent facial features and latent makeup features remains challenging.

3.3. Objective

To tackle this challenge, we introduce several losses to guide the decomposition. 1) The Perceptual loss is proposed to extract the facial features. 2) Intra-domain loss and inter-domain loss are designed for ensuring the after-makeup image and de-makeup image being at the makeup and non-makeup domain respectively. 3) Makeup loss is to extract the makeup features. 4) Cycle consistency loss

maintains the face and makeup information as input features. The details of loss functions and model architecture are presented as follows, while the effects of different losses are evaluated in Section 4.

Perceptual Loss. We first introduce the perceptual loss $\mathcal{L}_p$ to teach W how to extract facial features. Since the original source image is assumed to be non-makeup, we define the perceptual loss as follows.

$$\mathcal{L}_p = \|F_s^X - L_s^X\|_2, \quad (1)$$

which constrains the distance between the facial latent feature F_s^X and the latent features of original source image L_s^X .

Makeup Loss. Since the latent vector of the reference image includes makeup features, which means the W should be able to discriminate face features and makeup features. However, there is no image representing makeup styles, and thus the makeup features cannot be derived by transferring the image through Glow. Therefore, it is challenging to train how to describe a makeup style with several latent features. Here, we first tackle this problem by assuming that the latent features of a human face image are composed of facial features and makeup features. When the facial features

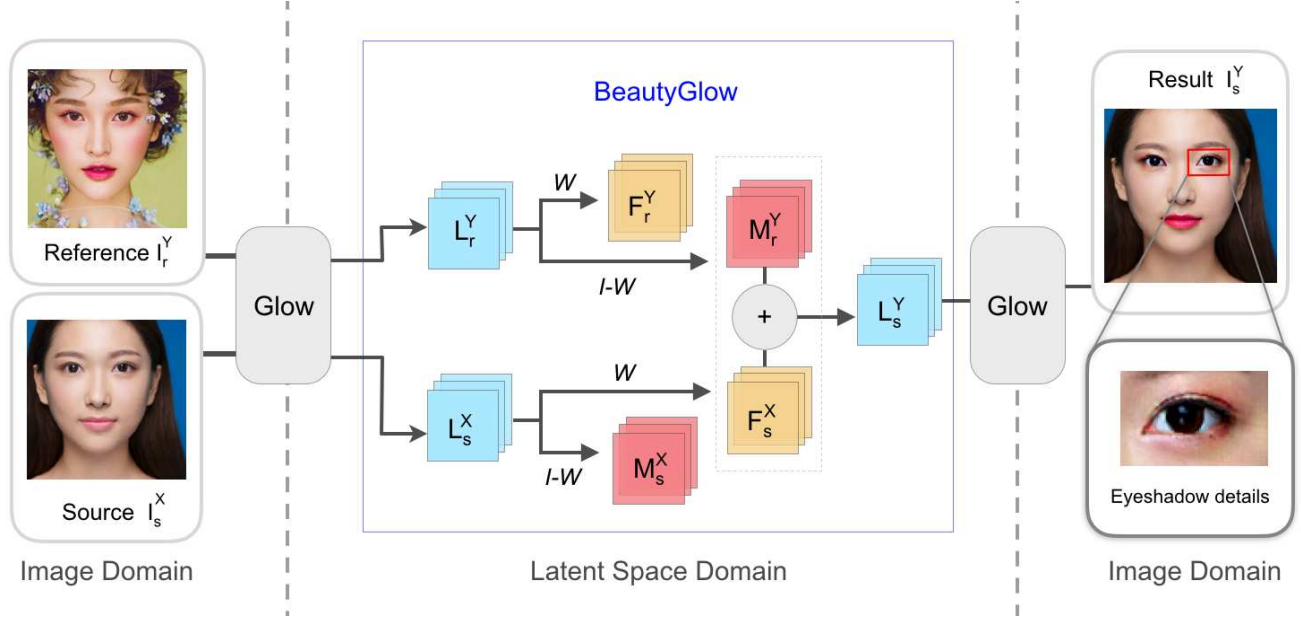


图2: The framework of the BeautyGlow. Glow transforms all images in makeup domain I^Y and all images in the non-makeup domain I^X as inputs: they are encoded into latent space Z , denoted as L^X and L^Y respectively. a vector obtained from L^X as source non-makeup, L_s^X , and a vector obtained from L^Y as reference makeup image, L_r^Y , are decomposed into two latent features (F_s^X, M_s^X) , (F_r^Y, M_r^Y) respectively through a transformation matrix W . L_s^Y is generated by adding F_s^X and M_r^Y together and L_s^Y can also be decomposed into reconstruction latent features, (F_s^Y, M_s^Y) .

通过 Glow[15]。

给定两张输入图像：源图 $I_s^X \in X$ 和参考图 $I_r^Y \in Y$ 分别被编码为两个潜在向量 $L_s^X \in Z$ 和 $L_r^Y \in Z$ ，使用的是Glow模型。随后，一个变换矩阵 W 用于提取面部特征表示为 $F_s^X = L_s^X \cdot W$ 以及妆容特征表示为 $M_r^Y = L_r^Y \cdot (I - W)$ ，其中 I 是单位矩阵。在此过程中，我们采用加法形式，即通过将 F_s^X 和 M_r^Y 相加来形成妆后潜在向量 $L_s^Y \in Z$ ，因为1) 加法形式可以即时计算；2) 加法形式中的插值直观易懂。接着，该妆后向量 L_s^Y 被解码回图像域作为妆后的图像 $I_s^Y \in Y$ 。BeautyGlow的模型架构如图2所示。

然而，如何学习转换矩阵 W 以将潜在向量完美分解为潜在面部特征和潜在妆容特征仍然具有挑战性。

3.3. 目的

为了应对这一挑战，我们引入了几种损失函数来指导分解。1) 感知损失用于提取面部特征。2) 域内损失和跨域损失设计用来确保化妆后的图像和卸妆后的图像分别处于化妆和非化妆的领域。3) 化妆损失用于提取化妆特征。4) 循环一致性损失

维持面部和妆容信息作为输入特征。损失函数的细节及模型架构如下所示，不同损失的效果将在第4节中进行评估。

感知损失。我们首先引入感知损失，用于教导模型如何提取面部特征。由于原始源图像是假定未化妆的，因此我们将感知损失定义如下。

$$\mathcal{L}_p = \|F_s^X - L_s^X\|_2, \quad (1)$$

这限制了面部潜在特征 F_s^X 与原始源图像的潜在特征 L_s^X 之间的距离。

补妆损失。由于参考图像的潜在向量包含化妆特征，这意味着 W 应该能够区分面部特征和化妆特征。然而，并没有代表特定化妆风格的图像，因此无法通过 Glow 转换图像来推导出化妆特征。因此，用几个潜在特征描述一种化妆风格变得具有挑战性。在这里，我们首先假设一个人脸图像的潜在特征由面部特征和化妆特征组成。当面部特征

are removed, the rest is makeup features. Meanwhile, as shown in Glow[15], attributes could be extracted through manipulating the latent features for images. Therefore, we first calculate the average latent vector of all images with makeup, denoted as $\bar{L}^Y$, and the average latent vector of all images without makeup, denoted as $\bar{L}^X$. Since the difference ($\bar{L}^Y - \bar{L}^X$) represents the direction from non-makeup latent vector to makeup latent vector, which should be similar to M_r^Y , we formulate the makeup loss $\mathcal{L}_m$ as follows.

$$\mathcal{L}_m = \|M_r^Y - (\bar{L}^Y - \bar{L}^X)\|_2. \quad (2)$$

Intra-Domain Loss. The makeup loss only forces M_r^Y to be similar with the direction from non-makeup latent vector to makeup latent vector, while the centroids of non-makeup latent vectors and makeup latent vectors are not fully exploited. The facial latent vectors of reference images, F_r^Y , are supposed to be close to non-makeup domain rather than makeup domain. Moreover, the after-makeup latent vectors (L_s^Y) are supposed to be close to the makeup domain instead of the non-makeup domain. Therefore, we use the average latent vector of all non-makeup images, $\bar{L}^X$, to represent the centroid of non-makeup domain, and the average latent vector of all makeup images, $\bar{L}^Y$ to represent the centroid of makeup domain. The intra-domain loss is then defined as

$$\mathcal{L}_{intra} = \|F_r^Y - \bar{L}^X\|_2 + \|L_s^Y - \bar{L}^Y\|_2. \quad (3)$$

Inter-Domain Loss. In addition to the intra-domain loss, we further introduce inter-domain loss to ensure that F_r^Y is away from the centroid of makeup domain to clearly decompose the facial latent vectors and makeup latent features effectively. Meanwhile, L_s^Y is also supposed to be away from the centroid of non-makeup domain. As such, we formulate the inter-domain loss as follows. Instead of the L2-norm, we calculate the similarity between two latent vectors A and B as

$$Sim(A, B) = \frac{sum(A \otimes B)}{|A||B|}, \quad (4)$$

where $\otimes$ denotes element-wise multiplication and $|\cdot|$ denote the L2-norm of the matrix. The loss function is expressed as

$$\mathcal{L}_{inter} = (1 + Sim(F_r^Y, \bar{L}^Y)) + (1 + Sim(L_s^Y, \bar{L}^X)) \quad (5)$$

Cycle Consistency Loss. In order to maintain the facial and makeup information, two cycle consistency losses are also designed in the latent space. Specifically, if we multiply the after-makeup latent vector, i.e., L_s^Y , with transformation matrix W , it is supposed to be close to the facial latent

vectors of the source image, i.e., F_s^Y . On the other hand, if we multiply L_s^Y with $(I - W)$, it is supposed to be close as makeup latent features of reference latent features M_r^Y . Therefore, the loss function is formulated as

$$\mathcal{L}_{cyc} = \|L_s^Y W - F_s^X\|_2 + \|L_s^Y (I - W) - M_r^Y\|_2 \quad (6)$$

Total loss. In sum, the overall loss function $\mathcal{L}$ is defined as follows.

$$\mathcal{L} = \mathcal{L}_p + \lambda_{cyc} \mathcal{L}_{cyc} + \lambda_m \mathcal{L}_m + \lambda_{ia} \mathcal{L}_{intra} + \lambda_{ie} \mathcal{L}_{inter} \quad (7)$$

where λ_{cyc} , λ_m , λ_{ia} , and λ_{ie} are the weights to control the relative importance of each term. The transformation matrix W can be trained via the objective function in Equation 7 with gradient descent based method, which will be detailed later.

4. Experimental Results

In this section, we first describe the implementation details and training parameter setting. Afterward, we briefly introduce the baselines, which are the state-of-the-art methods in makeup transfer. We conduct both qualitative and quantitative experiments to compare with baselines. A qualitative analysis on different losses of BeautyGlow is also presented. Finally, we show the advantages of flow-based method by presenting the makeup results from light to heavy, which is simply derived by interpolating the latent vectors and inverting back to image domain.

4.1. Implementation Details

Training Pipeline. We evaluate our method on MT dataset [19] consisting of about 4000 female images (1000 non-makeup images and 3000 makeup images). MT dataset includes different races, poses, expression and makeup styles varying from subtle to heavy. Therefore, we first follow the architecture of Glow [15] with pre-trained weights. Glow was originally trained on 5 machines with each 8 GPUs. Due to the limitation of the number of GPUs and GPU memories, we resize the training image size to 128x128. It takes about 3 days to train a Glow model with MT dataset.

Moreover, due to the resolution constraint, we first apply face parsing, i.e., separate different face parts and keep makeup details. Images are separated (with/without makeup) into left eye, right eye, lips, and the rest since eyes and lips are significantly changed by makeup. Please note that we only train single W which is shared by different parts of the faces. In post-processing, we use Poisson blending to integrate the generated makeup facial parts and the source image.

在去除相关特征后，剩余的即为妆容特征。如Glow[15]所示，可以通过操作潜在特征来提取属性。因此，我们首先计算所有带有妆容图像的平均潜在向量，记作 $\bar{L}^Y$ ，以及所有未带妆容图像的平均潜在向量，记作 $\bar{L}^X$ 。由于差值 $(\bar{L}^Y - \bar{L}^X)$ 表示从无妆潜在向量到有妆潜在向量的方向，这应该与 M_r^Y 相似，因此我们将妆容损失 $\mathcal{L}_m$ 定义如下。

$$\mathcal{L}_m = \|M_r^Y - (\bar{L}^Y - \bar{L}^X)\|_2. \quad (2)$$

域内损失。化妆损失仅强制要求 M_r^Y 与从非化妆潜在向量到化妆潜在向量的方向相似，而未充分利用非化妆和化妆潜在向量的中心点。参考图像的面部潜在向量 F_r^Y 应接近于非化妆领域而非化妆领域。此外，化妆后的潜在向量 L_s^Y 应接近于化妆领域而非非化妆领域。因此，我们使用所有非化妆图像的平均潜在向量 $\bar{L}^X$ 来表示非化妆领域的中心点，并用所有化妆图像的平均潜在向量 $\bar{L}^Y$ 来表示化妆领域的中心点。域内损失定义为

$$\mathcal{L}_{intra} = \|F_r^Y - \bar{L}^X\|_2 + \|L_s^Y - \bar{L}^Y\|_2. \quad (3)$$

跨域损失。除了域内损失之外，我们进一步引入跨域损失以确保 F_r^Y 远离化妆领域的中心点，从而清晰地分解面部潜在向量和化妆品特征。同时， L_s^Y 也应远离非化妆领域的中心点。因此，我们将跨域损失定义如下。我们不使用L2范数来计算两个潜在向量 A 和 B 之间的相似度，而是采取其他方法。

$$Sim(A, B) = \frac{sum(A \otimes B)}{|A||B|}, \quad (4)$$

其中 $\otimes$ 表示逐元素乘法， $|\cdot|$ 表示矩阵的L2范数。损失函数表示为

$$\mathcal{L}_{inter} = (1 + Sim(F_r^Y, \bar{L}^Y)) + (1 + Sim(L_s^Y, \bar{L}^X)) \quad (5)$$

周期一致性损失。为了保持面部和妆容信息，在潜在空间中设计了两个周期一致性损失。具体而言，如果我们将化妆后的潜在向量，即 L_s^Y ，与变换矩阵 W 相乘，则应接近于面部的潜在向量。

源图像的向量，即 F_s^Y 。另一方面，若将 L_s^Y 与 $(I - W)$ 相乘，则应接近参考潜在特征 M_r^Y 的妆容潜在特征。因此，损失函数被定义为

$$\mathcal{L}_{cyc} = \|L_s^Y W - F_s^X\|_2 + \|L_s^Y (I - W) - M_r^Y\|_2 \quad (6)$$

总损失。综上所述，整体损失函数 $\mathcal{L}$ 定义如下。

$$\mathcal{L} = \mathcal{L}_p + \lambda_{cyc} \mathcal{L}_{cyc} + \lambda_m \mathcal{L}_m + \lambda_{ia} \mathcal{L}_{intra} + \lambda_{ie} \mathcal{L}_{inter} \quad (7)$$

其中， λ_{cyc} 、 λ_m 、 λ_{ia} 和 λ_{ie} 是用于控制每一项相对重要性的权重。可以通过基于梯度下降法的Equation 7目标函数来训练变换矩阵 W ，具体细节将在后面详细说明。

4. 实验结果

在本节中，我们首先描述实现细节和训练参数设置。随后，我们将简要介绍基准方法，这些是当前最先进的化妆迁移技术。我们进行了定性和定量实验以与基准进行比较。还对BeautyGlow的不同损失函数进行了定性分析。最后，通过插值隐变量并逆向映射回图像域来展示从轻到重的妆容结果，以此展示了基于流的方法的优势。

4.1 实现细节

训练管道。我们在包含约4000张女性图像（其中1000张为未化妆图像，3000张为化妆图像）的MT数据集[19]上评估我们的方法。该数据集包括不同种族、姿态、表情和从细微到浓重的各种妆容。因此，我们首先遵循Glow [15]的架构，并使用预训练权重。Glow最初是在配备了8个GPU的5台机器上进行训练的。由于受制于可用GPU数量及内存限制，我们将训练图像尺寸调整为128x128。使用MT数据集训练一个Glow模型大约需要3天时间。

此外，由于分辨率限制，我们首先进行面部解析，即分离不同的面部部分并保留妆容细节。图像被分割为左眼、右眼和嘴唇（有无妆容），因为眼睛和嘴唇在化妆后变化显著。请注意，我们仅训练单一的 W 模型，该模型由不同面部区域共享。在后期处理中，我们使用泊松融合技术将生成的带妆面部部分与源图像进行整合。

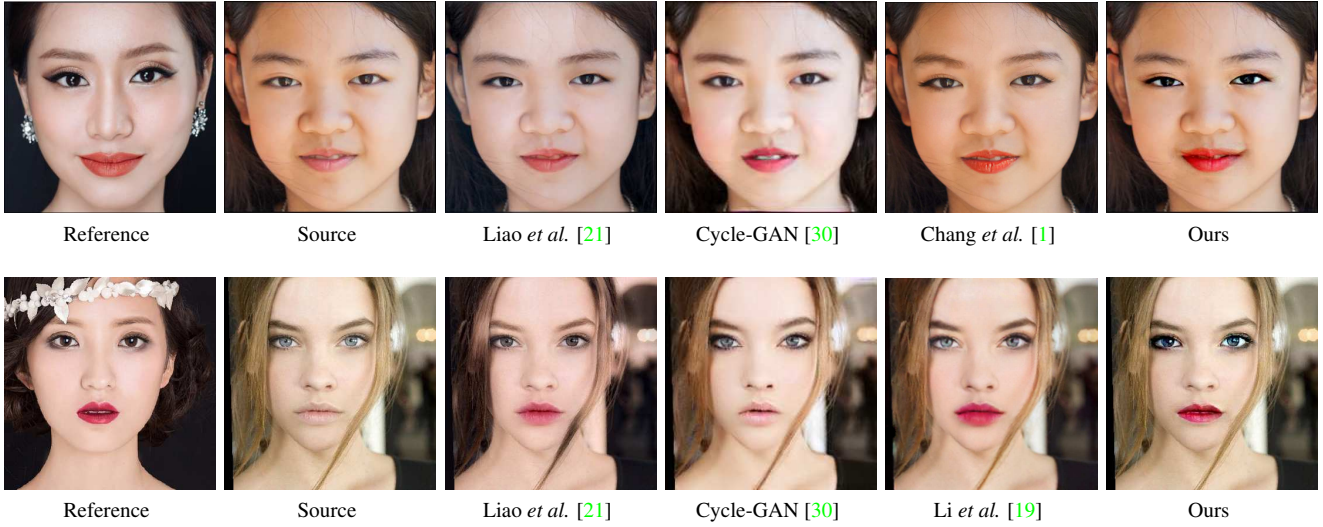


Figure 3: Makeup Transfer Results

Training Details. Given the dataset of unpaired makeup and non-makeup images, we first use Glow to transform the images into the latent space and calculate the centroids of makeup and non-makeup latent domains. Afterward, we train the transformation matrix W via the objective function in Equation 7 with Adam optimizer, which is a classic extension of stochastic gradient descent procedure to update model weights iteratively based on the training data. The learning rate is set as 0.001 with the batch size as 100. The size of the transformation matrix W is 128×128 , while the control parameters of perceptual loss, makeup loss, intra-domain loss, inter-domain loss, and cycle consistency loss are set as ($\lambda_p = 0.01$, $\lambda_{cyc} = 0.001$, $\lambda_m = 0.1$, $\lambda_{intra} = 0.1$, $\lambda_{inter} = 1000$), respectively.

4.2. Baselines

BeautyGlow is compared with several methods including traditional method, style transfer and image domain adoption via qualitative and quantitative experiments as listed below.

Deep Image Analogy [21] extracts features by CNN and adapts the notion of image analogy to the deep feature space for obtaining semantically-meaningful dense correspondences. Here, we add the WLS filter to keep details for photo-to-photo transfer.

CycleGAN [30] is an unsupervised image-to-image translation work, which does not require paired images. Therefore, makeup images and non-makeup images are regarded as two different domains for training.

PairedCycleGAN [1] is a paired image-to-image makeup transfer method which aims to transfer a specific reference makeup to a source image.

BeautyGAN [19] is an instance-level makeup transfer method for makeup transfer and removal via GAN. The results are enhanced by matching the color histogram in different segments of the face.

4.3. Qualitative Comparison

In Figure 3, we demonstrate a qualitative comparison of the baselines and our results. The results produced by Image Analogy show that eyeshadows and eyeliners disappear and only subtle color changes stay around the eyes. This is because it applies whole image transformation, parts aside from makeup including skin tone and hair might also be changed to an unnatural color. Compared with Image Analogy, image domain adoption methods can create a more realistic look. CycleGAN [30] only synthesizes a general makeup on the non-makeup face. The makeup style or amount cannot be adjusted since the images are transferred in only two general domain, makeup and non-makeup. The general makeup, though realistic, cannot be used for specific makeup trials. Moreover, PairedCycleGAN [1] can transfer makeup realistically and correctly in pairs. However, the disadvantages of PairedCycleGAN is that makeup are generated by a fixed adversarial network so that users cannot adjust the makeup to be lighter or heavier. Therefore, when transferring some heavy makeup, the results may look unpleasant and sometimes unrealistic, but users can only find new reference images with lighter makeup, which is time-consuming and even unfeasible.

Comparing to the baselines, our method successfully decomposes the makeup latent vectors and non-makeup latent vectors, so it does not change the look of the image severely like style transfer. Meanwhile, we can generate a realistic

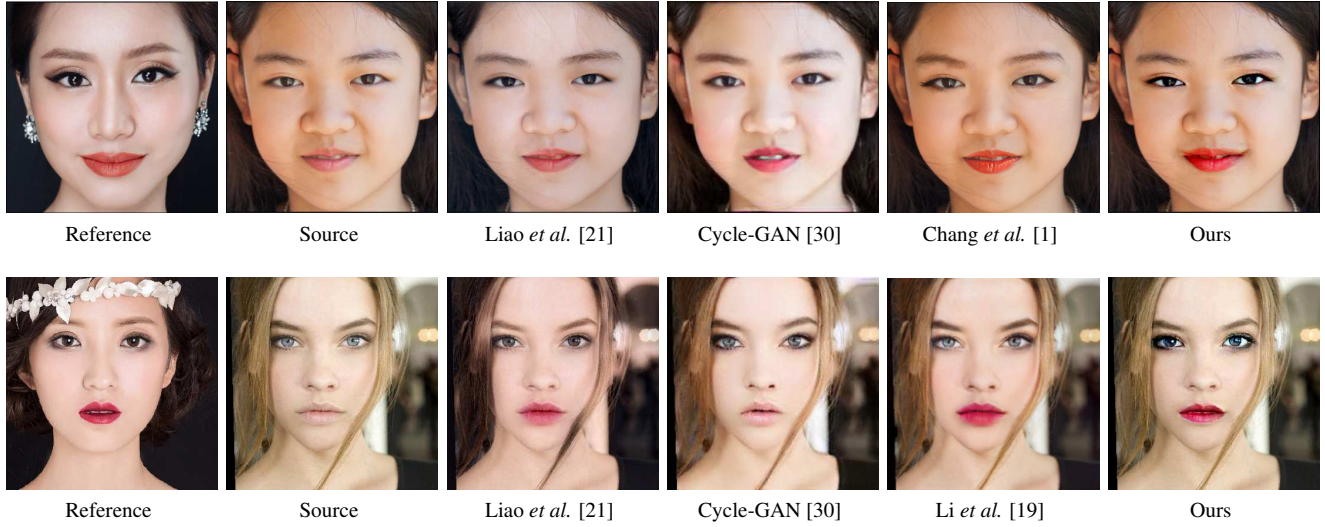


图3: 妆容迁移结果

训练细节。给定未配对的化妆和非化妆图像数据集，我们首先使用Glow将图像转换到潜在空间，并计算化妆和非化妆潜在域的中心点。随后，通过方程7中的目标函数，利用Adam优化器训练变换矩阵 W ，该优化器是随机梯度下降过程的经典扩展，基于训练数据迭代更新模型权重。学习率为0.001，批处理大小为100。变换矩阵 W 的尺寸为 128×128 ，而感知损失、化妆损失、域内损失、域间损失和循环一致性损失的控制参数分别设置为($\lambda_p = 0.01, \lambda_{cyc} = 0.001, \lambda_m = 0.1, \lambda_{intra} = 0.1, \lambda_{inter} = 1000$)。

4.2. 基线方案

BeautyGlow 通过定性和定量实验与几种方法进行了比较，包括传统方法、风格迁移和图像域适应。

深度图像类比法[21]通过CNN提取特征，并将图像类比的概念应用于深层特征空间，以获取语义上有意义的密集对应关系。在这里，我们添加了WLS滤波器以保留细节，以便于照片到照片的转换。

CycleGAN[30]是一种无监督的图像到图像翻译工作，不需要成对的图像。因此，化妆图像和未化妆图像被视为两个不同的训练域。

配对CycleGAN[1]是一种配对图像到图像的化妆转移方法，旨在将特定的参考妆容转移到源图像上。

BeautyGAN[19]是一种通过生成对抗网络（GAN）实现化妆转移和卸妆的实例级方法。该方法通过匹配面部不同区域的颜色直方图来增强结果。

4.3. 定性比较

图3展示了基线方法与我们结果的定性比较。Image Analogy 方法产生的结果显示眼影和眼线消失，只有细微的颜色变化留在眼睛周围。这是因为该方法应用了整个图像变换，除了妆容之外的部分（如肤色和头发）也可能被改变成不自然的颜色。相比之下，基于图像域的方法可以创建更逼真的效果。CycleGAN [30] 只在非化妆脸上合成一般的妆容。由于图像仅在两个通用领域之间转换（即化妆与非化妆），因此无法调整妆容的风格或程度。虽然这种一般化的妆容看起来很真实，但不能用于特定的试妆场景。此外，PairedCycleGAN [1] 可以成对地逼真且准确地转移妆容。然而，PairedCycleGAN 的缺点是生成的妆容由固定的对抗网络决定，用户无法调整妆容的浓淡程度。因此，在转换一些重妆时，结果可能看起来不悦目甚至不真实，而用户只能寻找新的参考图片以找到较轻的妆容，这既耗时又有时根本不可行。

与基线方法相比，我们的方法成功地分解了化妆隐变量和未化妆隐变量，因此不会像风格迁移那样严重改变图像的外观。同时，我们可以生成逼真的图像。

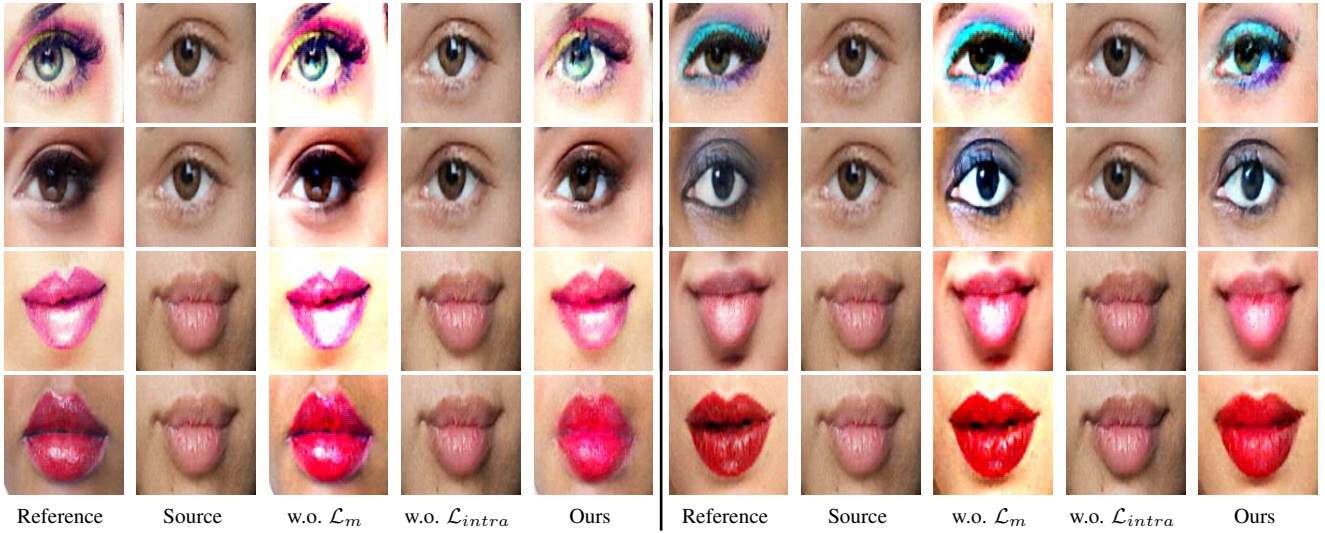


Figure 4: Analysis of Different Loss Terms of BeautyGlow.

Table 1: User Study Results

preference comparison	our result	baseline
Ours / Liao <i>et al.</i> [21]	60%	40%
Ours / Li <i>et al.</i> [19]	55%	45%
Ours / Chang <i>et al.</i> [1]	45%	55%

and accurate makeup for a non-makeup face comparable to PairedCycleGAN. In Section 4.6, we show the power of the ability to fine-tune the amount of makeup and customize it for each face to get the perfect result.

4.4. Quantitative Comparison

For quantitative evaluation of BeautyGlow, a user study is conducted with 50 volunteers (34 males and 16 females) aged from 18 years old to 35 years old. To simplify the process of comparison, a standard A/B test is used. Therefore, each user is asked to compare the results of BeautyGlow with Image Analogy [21], BeautyGlow with PairedCycleGAN [1], and BeautyGlow with BeautyGAN [19]. For each user, we randomly choose fifteen pairs of source and reference images and generate the results by different methods. The positions of different methods for the A/B test is also randomly selected to avoid the bias. Table 1 shows the results of the A/B test. For the ease of comparison, we normalized the votes and get the preference percentage. The results show that BeautyGlow significantly outperforms Image Analogy while is comparable to BeautyGAN and PairedCycleGAN. Note that we observe some unusual situations that user preference decreases while transferring heavy makeup. Some exaggerated makeup is not suitable for the source image, forcing users to choose a rather subtle

makeup face. As a result, with the latent space manipulation in our model, we can fine-tune the makeup amount for a more preferred look.

4.5. Loss Analysis

We introduce five losses for training the transformation function. In Figure 4, we show the results of two important loss terms by training the transformation function W without the loss term. When removing $\mathcal{L}_m$, the transformation matrix learns makeup features without average makeup features. Hence, the makeup style is over-enhanced, leading to unnatural after-makeup images, especially in the regions of eyes. On the other hand, without $\mathcal{L}_{intra}$, the after-makeup image contains no makeup components because there is no loss term constraining the resulting image to be close to the centroid of the makeup images. In addition, facial features are supposed to be at the group of non-makeup images, so the facial features extracted from makeup images would include makeup features.

We show the results of face components like eyes and mouth for different pairs of a reference image and a source image. In the first two rows, without using makeup loss, the makeup style is over-emphasized and the color of other regions is changed dramatically. When the intra-loss term is removed, the result looks like the same as the source image. Moreover, in the second row, the color of the results without makeup loss is obviously brighter than the source and the reference image.

4.6. Latent Space Manipulation

The highlighted property of BeautyGlow is the ability to manipulate the latent vectors for generating the same makeup style with different lightness. To show the after-

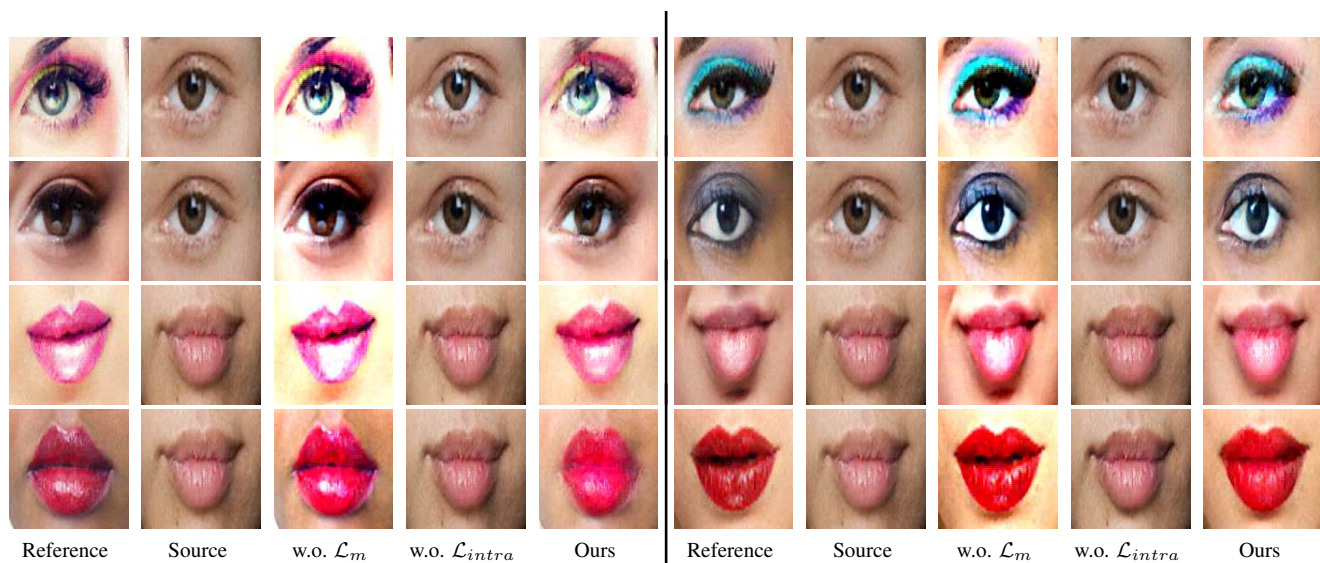


图4: BeautyGlow不同损失项的分析。

表1: 用户研究结果

preference comparison	our result	baseline
Ours / Liao <i>et al.</i> [21]	60%	40%
Ours / Li <i>et al.</i> [19]	55%	45%
Ours / Chang <i>et al.</i> [1]	45%	55%

与 PairedCycleGAN 相当，能够为未化妆的脸生成逼真且准确的妆容。在第 4.6 节中，我们展示了微调化妆量并针对每张脸进行定制以获得完美效果的能力的强大之处。

4.4. 定量比较

为了对BeautyGlow进行定量评估，我们开展了一项包含50名志愿者（34名男性和16名女性）的用户研究，年龄范围从18岁到35岁。为简化比较过程，采用标准A/B测试方法。因此，每位参与者需要将BeautyGlow与Image Analogy [21]、BeautyGlow与PairedCycleGAN [1]以及BeautyGlow与BeautyGAN [19]的结果进行对比。对于每个用户，我们随机选取了十五组源图像和参考图像，并通过不同方法生成结果。A/B测试中各方法的位置也是随机选择的，以避免偏差。表1显示了A/B测试的结果。为了便于比较，我们将投票数进行了标准化处理并计算出偏好百分比。结果显示，BeautyGlow显著优于Image Analogy，而与BeautyGAN和PairedCycleGAN相当。值得注意的是，在进行浓妆转换时，我们观察到一些不寻常的情况：用户的偏好会下降，因为某些夸张的化妆效果并不适合源图像，迫使用户选择较为温和的效果。

通过调整潜空间，我们的模型可以微调化妆量，以达到更理想的妆容效果。

4.5. 延迟分析

我们介绍了五种用于训练转换函数的损失函数。在图4中，展示了通过去除某个损失项来训练转换函数 W 时两个重要损失项的结果。当移除 $\mathcal{L}_m$ 后，转换矩阵学习到的是化妆特征而非平均化妆特征，因此妆容被过度增强，导致化完妆后的图像显得不自然，尤其是在眼部区域。另一方面，如果没有 $\mathcal{L}_{intra}$ ，化完妆的图像将不含任何化妆成分，因为没有损失项来约束生成的图像接近于所有化妆图像的中心点。此外，面部特征应位于非化妆图像组中，因此从化妆图像提取出的面部特征会包含化妆特征。

我们展示了不同参考图像和源图像对中人脸组件（如眼睛和嘴巴）的结果。在前两行中，不使用化妆损失时，妆容风格被过度强调，并且其他区域的颜色发生了显著变化。当去掉内部损失项后，结果看起来与源图像相同。此外，在第二行中，没有使用化妆损失的情况下，结果的颜色明显比源图像和参考图像更亮。

4.6. 隐空间操纵

BeautyGlow 的突出特性是能够操纵潜在向量，生成不同亮度下的正常妆容风格。为了展示其效果，在…

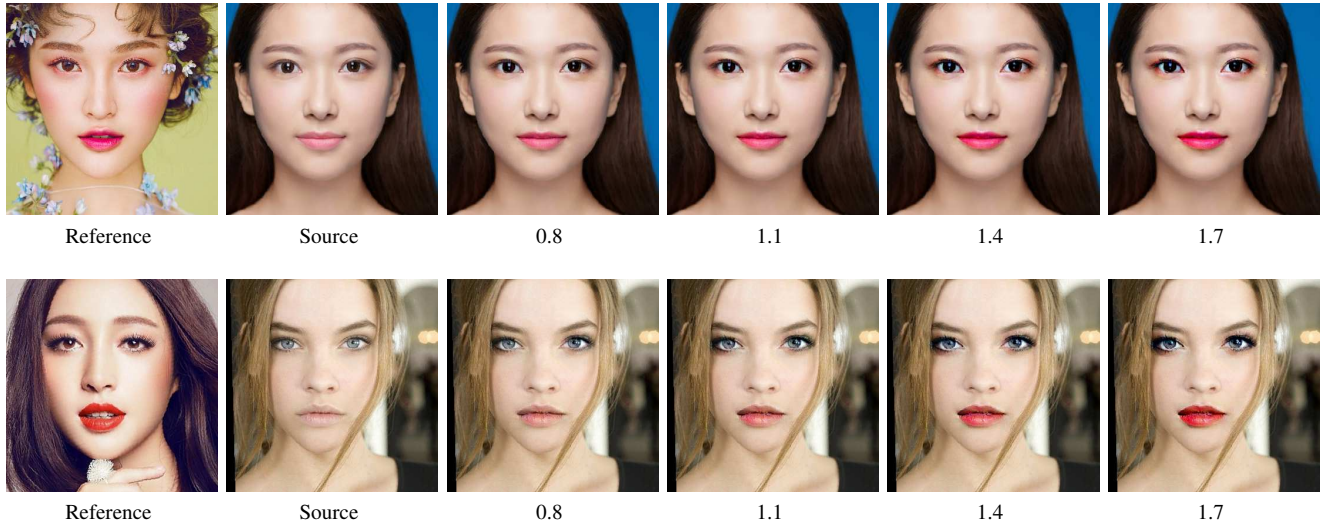


Figure 5: The obviousness of the makeup

makeup face with different makeup lightness, we manipulate the weight of the makeup features in the latent space. In Figure 5, we synthesize 3 different lightness of two makeup styles on different faces with the magnification of 0.7, 1.0, 1.5. In the first row, as the magnification increases, the eyeshadows and lip color become redder and redder. In the second row, the eyeshadows also become darker and the lip color becomes redder. It is worth noting that even though the makeup is magnified, the face is still not distorted, which means the transformation matrix completely decomposes the latent vector into the facial features and makeup features for the manipulations.

5. Conclusions and Future Work

As quoted from Estée Lauder, “Glow is the essence of beauty,” which is also true for makeup transfer. In this paper, we propose BeautyGlow to decompose the latent vectors derived from the Glow model into makeup and non-makeup latent vectors. Since there is no paired dataset, we formulate a new loss function containing perceptual loss, makeup loss, inter-domain loss, intra-domain loss, and cycle consistency loss, to guide the optimization of the transformation matrix. Experimental results show that the makeup transfer quality of BeautyGlow is comparable to the state-of-the-art methods, while the unique ability to manipulate latent vectors allows BeautyGlow to realize on-demand makeup transfer. In the future, we plan to apply the proposed general framework to other image synthesis applications.

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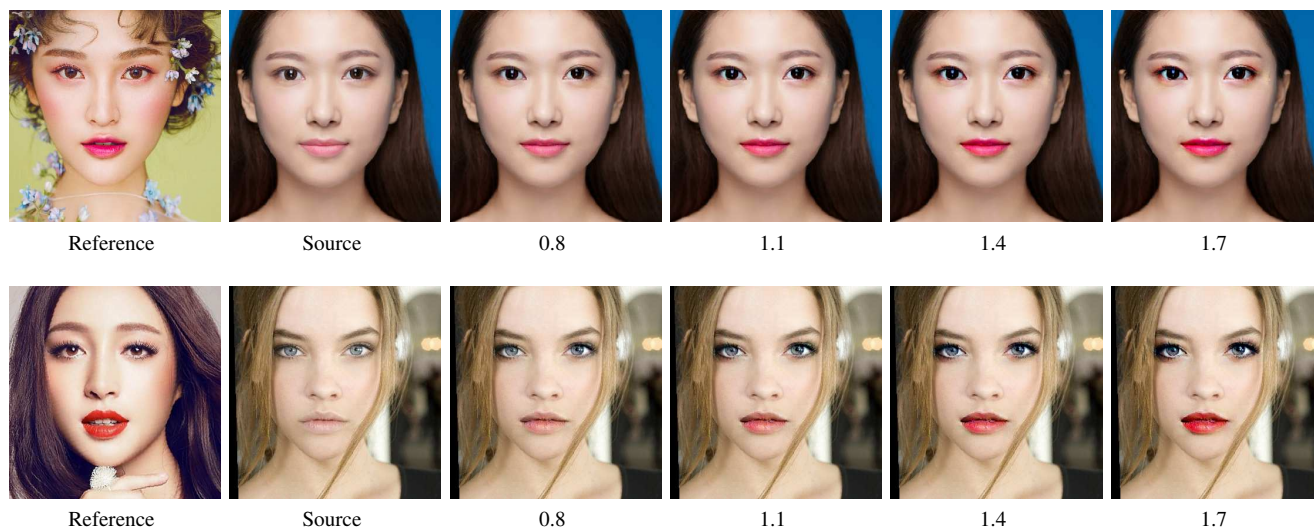


图5：妆容的明显性

针对不同妆容的浓淡，我们在潜在空间中调整妆容特征的权重。如图5所示，我们以0.7、1.0和1.5的不同放大倍数，在不同的脸上合成了两种妆容风格的三种不同浓淡效果。第一行中，随着放大的增加，眼影和唇色逐渐变红。第二行中，眼影变得更深，唇色也更红。值得注意的是，即使妆容被放大了，人脸也没有变形，这意味着变换矩阵完全将潜在向量分解为人脸特征和妆容特征以进行调整。

5. 结论与未来工作

正如Estée Lauder所言，“光泽是美的精髓”，这一点同样适用于化妆品迁移。在本文中，我们提出BeautyGlow方法，将从Glow模型导出的潜在向量分解为化妆和非化妆潜在向量。由于缺乏配对数据集，我们设计了一个新的损失函数，包含感知损失、化妆品损失、跨域损失、同域损失以及循环一致性损失，以指导变换矩阵的优化。实验结果表明，BeautyGlow在化妆品迁移质量上与现有最佳方法相当，而其独特的能力在于可以操作潜在向量，从而实现按需化妆品迁移。未来，我们计划将提出的通用框架应用于其他图像合成应用中。

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